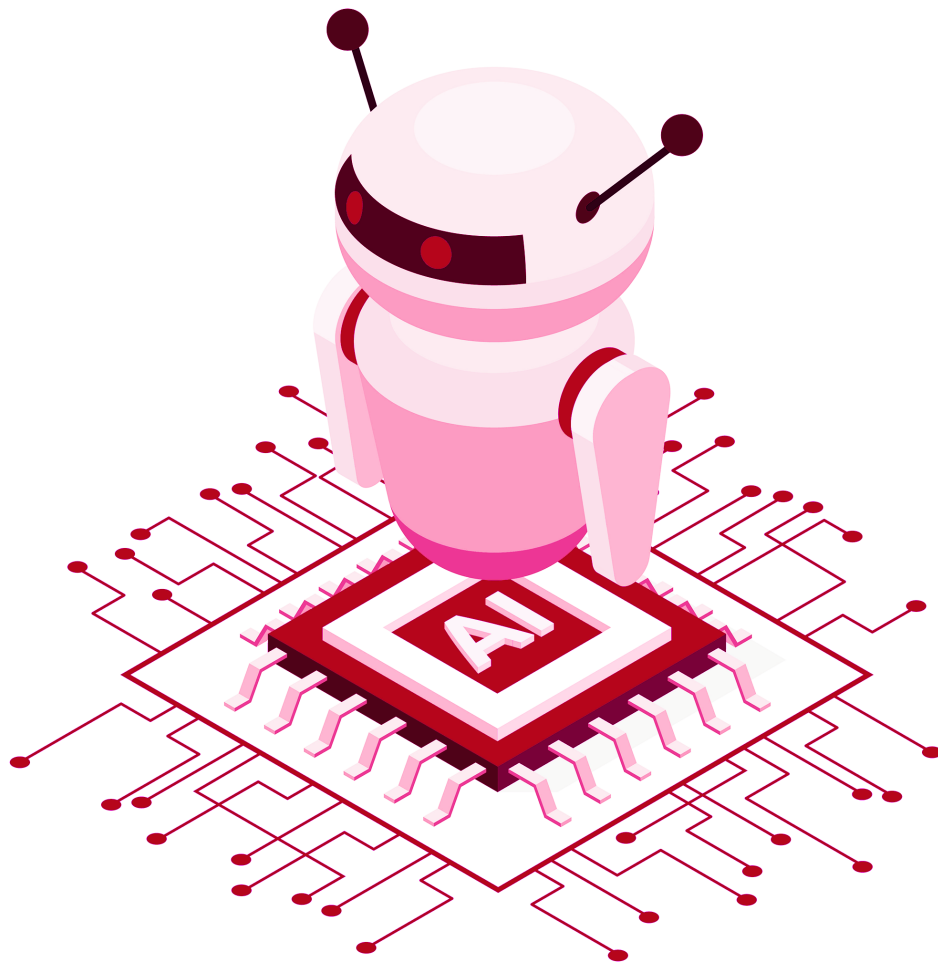




Deep Reinforcement Learning

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Solution for Homework 12:

Offline Methods

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1 Part 1 [60-points]

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1. Considering the Bellman update, explain with reasoning why value estimation suffers from overestimation in the offline framework. [10-points]

$$Q(s, a) \leftarrow r(s, a) + \mathbb{E}_{a' \sim \pi_{\text{new}}} [Q(s', a')]$$

Answer:

The Bellman update $Q(s, a) \leftarrow r(s, a) + \mathbb{E}_{a' \sim \pi_{\text{new}}} [Q(s', a')]$ suffers from overestimation in offline RL due to Extrapolation error. The expectation over a' is computed using actions not present in the dataset. When π_{new} selects OOD (out-of-distribution) actions with erroneously high Q -values (due to lack of data constraints), these overestimates propagate backward.

2. One of the solutions to address the overestimation problem in the offline framework is CQL, whose objective function for computing the value is given below. Explain the role of each of the four terms in this objective function. [20-points]

$$\begin{aligned} \hat{Q}^T = \arg \min_Q \max_{\mu} & \alpha \mathbb{E}_{s \sim D, a \sim \mu(a|s)} [Q(s, a)] \\ & - \alpha \mathbb{E}_{(s,a) \sim D} [Q(s, a)] \\ & - \mathbb{E}_{s \sim D} [\mathcal{H}(\mu(\cdot|s))] \\ & + \mathbb{E}_{(s,a,s') \sim D} [(Q(s, a) - (r(s, a) + \mathbb{E}[Q(s', a')]))^2] \end{aligned}$$

Answer:

- (a) $\alpha \mathbb{E}_{s \sim D, a \sim \mu} [Q(s, a)]$: Penalizes OOD actions by maximizing Q under adversarially chosen μ to identify overestimated actions.
 - (b) $-\alpha \mathbb{E}_{(s,a) \sim D} [Q(s, a)]$: Preserves dataset values by minimizing Q on in-distribution (s, a) pairs.
 - (c) $-\mathbb{E}_s [\mathcal{H}(\mu)]$: Regularizes μ by encouraging high entropy to prevent adversarial collapse to a single action.
 - (d) Bellman error term: Enforces consistency with the Bellman equation on dataset transitions.
3. Rewrite the optimization problem from part 3 as a minimization-only problem. [20-points]

Answer:

Solve the inner maximization over μ :

$$\max_{\mu} \{ \alpha \mathbb{E}_{a \sim \mu(\cdot|s)} [Q(s, a)] + \mathcal{H}(\mu(\cdot|s)) \}$$

For a fixed s , we solve:

$$\max_{\mu(\cdot|s)} \left(\sum_a \mu(a|s) Q(s, a) - \sum_a \mu(a|s) \log \mu(a|s) \right)$$

subject to $\sum_a \mu(a|s) = 1$. Introduce the Lagrangian:

$$\mathcal{L} = \sum_a \mu(a|s) Q(s, a) - \sum_a \mu(a|s) \log \mu(a|s) + \lambda \left(1 - \sum_a \mu(a|s) \right).$$

Taking the derivative with respect to $\mu(a|s)$ and setting it to zero:

$$\frac{\partial \mathcal{L}}{\partial \mu(a|s)} = Q(s, a) - \log \mu(a|s) - 1 - \lambda = 0.$$

Solving for $\mu(a|s)$:

$$\log \mu(a|s) = Q(s, a) - 1 - \lambda \implies \mu(a|s) = \exp(Q(s, a) - 1 - \lambda).$$

Normalize to satisfy $\sum_a \mu(a|s) = 1$:

$$\begin{aligned} \sum_a \exp(Q(s, a) - 1 - \lambda) &= 1 \\ \implies \exp(-1 - \lambda) \sum_a \exp(Q(s, a)) &= 1 \\ \implies \exp(-1 - \lambda) &= \frac{1}{\sum_a \exp(Q(s, a))}. \end{aligned}$$

Thus, the optimal $\mu^*(a|s)$ is:

$$\mu^*(a|s) = \frac{\exp(Q(s, a))}{\sum_{a'} \exp(Q(s, a'))}.$$

Substitute μ^* back into the objective for a single state s :

$$\mathbb{E}_{a \sim \mu^*(\cdot|s)}[Q(s, a)] + \mathcal{H}(\mu^*(\cdot|s)) = \sum_a \mu^*(a|s) Q(s, a) - \sum_a \mu^*(a|s) \log \mu^*(a|s).$$

Using $\mu^*(a|s) = \frac{\exp(Q(s, a))}{Z(s)}$ where $Z(s) = \sum_a \exp(Q(s, a))$:

$$\begin{aligned} &\sum_a \frac{\exp(Q(s, a))}{Z(s)} Q(s, a) - \sum_a \frac{\exp(Q(s, a))}{Z(s)} \log \left(\frac{\exp(Q(s, a))}{Z(s)} \right) \\ &= \sum_a \frac{\exp(Q(s, a))}{Z(s)} Q(s, a) - \sum_a \frac{\exp(Q(s, a))}{Z(s)} (Q(s, a) - \log Z(s)) \end{aligned}$$

Simplify:

$$\begin{aligned} &= \sum_a \frac{\exp(Q(s, a))}{Z(s)} Q(s, a) - \sum_a \frac{\exp(Q(s, a))}{Z(s)} Q(s, a) \\ &\quad + \log Z(s) \sum_a \frac{\exp(Q(s, a))}{Z(s)} \\ &= 0 + \log Z(s) \cdot 1 \\ &= \log Z(s) \end{aligned}$$

Thus, the maximum value is $\log \sum_a \exp(Q(s, a))$.

The inner maximization (over states) yields:

$$\alpha \mathbb{E}_{s \sim \mathcal{D}} \left[\log \sum_a \exp(Q(s, a)) \right] - \alpha \mathbb{E}_{s, a \sim \mathcal{D}} [Q(s, a)].$$

The minimization objective is:

$$\begin{aligned} \hat{Q} = \arg \min_Q & \alpha \mathbb{E}_{s \sim \mathcal{D}} \left[\log \sum_a \exp(\alpha Q(s, a)) \right] \\ & - \alpha \mathbb{E}_{(s, a) \sim \mathcal{D}} [Q(s, a)] \\ & + \mathbb{E}_{(s, a, s') \sim \mathcal{D}} [(Q(s, a) - (r + \gamma Q(s', a')))^2] \end{aligned}$$

4. To apply this method in model-based reinforcement learning, what changes are needed in the objective function? Rewrite the new objective function. [10-points]

Answer:

In model-based RL, replace the Bellman error term with model-generated transitions:

$$\begin{aligned} \hat{Q} = \arg \min_Q & \alpha \mathbb{E}_{s \sim \mathcal{D}} \left[\log \sum_a \exp(\alpha Q(s, a)) \right] \\ & - \alpha \mathbb{E}_{(s, a) \sim \mathcal{D}} [Q(s, a)] \\ & + \mathbb{E}_{(s, a) \sim \mathcal{D}} \mathbb{E}_{s' \sim T_\phi(\cdot | s, a)} [(Q(s, a) - (r + \gamma Q(s', a')))^2] \end{aligned}$$

where T_ϕ is the learned dynamics model. The conservative regularization (first two terms) remains on the real dataset states.

References

[1] [Cover image designed by freepik](#)